Correct-Response Novelty in Full-Weight LLM Evolution

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Abstract

We test an adaptation of historical response novelty to full-weight Gaussian evolution strategies on witnessed arithmetic problems. The comparison uses one paired training seed and the original pretrained instruction model without task-specific updates here. The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES. The paired interval includes zero, so this does not establish a performance difference. Problem-resampling intervals exclude training-seed variability. All tables are generated from persisted evaluations; incomplete outcomes are labeled.

Context, motivation and introduction

The experiment asks whether exploration among correct answer forms improves held-out task success. ExTra [1] supplies the historical semantic novelty concept; ES at Scale [2] supplies the full-weight optimization reference. Their combination uses population-level ES rather than GRPO, with a different sibling population and task, so it is an adaptation rather than a reproduction. A reward-only ES control is necessary because ES can itself change solution coverage [3].

Background and related work

Novelty-seeking ES was previously studied in control [4]. LLM-specific alternatives include EVOL-RL's within-group novelty [5], DARLING's quality/diversity objective [6], and the model/task coevolution of AC/DC [7]. These use different training or population mechanisms. The literature notes distinguish prompt evolution, artifact search and model-parameter training. Only explicitly identified dependencies have pinned code/license checks; peripheral references are not implementation validations.

Method

The model is Qwen/Qwen2.5-0.5B-Instruct, revision 7ae557604adf67be50417f59c2c2f167def9a775. Each generation evaluates 30 independent Gaussian candidates (sigma 0.001) on 32 rotating training problems and 16 fixed training probes. Population-z-scored task reward is combined with 1.0 times correctness-gated cosine novelty for the main condition. The archive retains 6 correct embeddings per probe after seeded insertion. These coefficients weight noise directly, without another normalization; effective alpha is 0.0005. FP32 masters reconstruct BF16 candidates afresh and tied tensors are updated once. Prefix regeneration and backpropagation are absent.

Qwen/Qwen2.5-0.5B-Instruct: 24 layers, width 896, 14 query heads and 2 key/value heads; 494,032,768 unique parameter elements in 290 tensors. Architecture comes from the exported config; counts come from its safetensors headers. This does not establish historical generation-zero tensor equality.

For candidate i, Q_i is the mean response reward: exact success receives 1, a legal expression using every supplied number but missing the target receives 0.1, and an invalid expression receives 0. For probe p, n_i,p = I(correct_i,p) [1 - max cosine(e_i,p, z)]. The reference set consists of correct historical responses and correct responses from other candidates on that same probe; self-comparisons are excluded. An empty set has maximum zero. N_i is the probe mean. History is frozen while scoring the whole generation, then correct embeddings are appended in seeded shuffled order and the oldest entries are removed.

Z_i = (Q_i - mean(Q)) / (std_population(Q) + 1e-8), with zero Z for constant Q. W_i = Z_i + 1.0 N_i for the main arm and W_i = Z_i for control. The update is theta_next = theta + (alpha / population) sum_i W_i epsilon_i. Each epsilon_i is independent standard Gaussian noise over every unique parameter tensor. A common positive novelty bonus is retained. Alpha includes the inverse-noise factor; it is not an additional alpha/sigma update. MiniLM is frozen, and both conditions compute the same categories of probe work. Complete response strings are passed to MiniLM, whose pinned sentence_bert_config.json caps inputs at 256 WordPiece tokens including special tokens [9]. This is not a guarantee that every response token contributes to novelty.

Protocol amendment A replaced combined-reward ranks with source-aligned z-scored quality plus post-normalization novelty. It was published after calibration and partial original-model evaluation were visible, but before either trained condition began. The run records describe both arms as starting from the original pretrained weights; their generation-zero tensors were not

retained, so equality cannot now be independently re-established. Recorded coefficients follow the amended rule. The original registration, changed configuration, code revisions and unchanged absolute deadline are retained in the run record. This ordering follows ES at Scale Algorithm 2 and ExTra Equation 2; it is not evidence that the hybrid outperforms the abandoned rule.

Data, evaluation and preregistered analysis

Synthetic problems have exact rational witnesses. Global number-multiset isolation prevents permutations or alternative targets crossing splits. Only supplied integer literals and binary +, -, *, / in Python expression syntax are accepted [10]. The grader accepts either a bare expression or the contents of one answer tag (surrounding prose is ignored). Unary signs, powers, floor division, extra literals and unparenthesized line breaks are rejected. Greedy accuracy on the designated primary split at the final scheduled checkpoint is primary. The directional hypothesis is falsified for this pilot if main minus control accuracy is nonpositive. A paired problem bootstrap quantifies conditional uncertainty. Sampled pass@k is the exact uniform-subset success probability; unbiasedness for model draws assumes independent samples [8]. Validation cannot select checkpoints. Exact dataset hashes, schedules, timing rules and sampling settings are recorded in reports/preregistration.md and configs/pilot.json.

Dataset counts: train 8192, validation 512, test 2048, hard 1024. Greedy responses allow at most 128 new tokens. The sampled evaluation uses 16 responses on each of 256 fixed original-test problems (never the fresh split), with temperature 1.0, top-p 1.0 and top-k disabled. pass@k = 1 - C(n-c,k)/C(n,k), where c is the number of correct samples. Intervals use 10000 paired resamples, seed 20261107.

Results and claims

The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES. The paired interval includes zero, so this does not establish a performance difference.

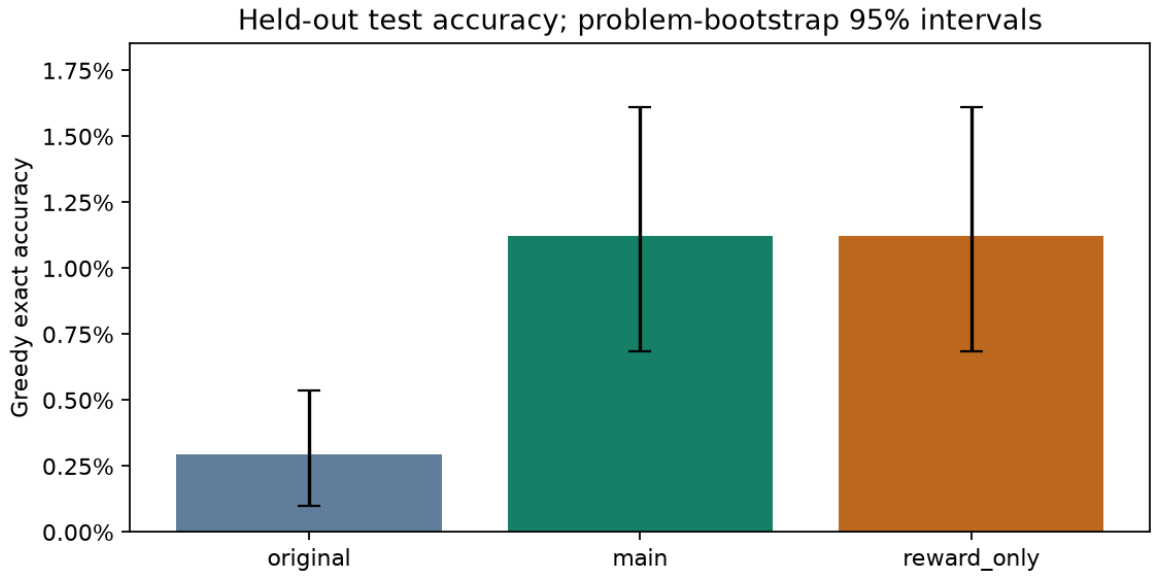
Condition	Test greedy accuracy	95% interval	Hard greedy accuracy	Original test pass@1	Original test pass@8	Status
original	0.29%	0.10% - 0.54%	0.10%	0.12%	0.89%	complete
main	1.12%	0.68% - 1.61%	0.00%	0.27%	2.06%	complete
reward_only	1.12%	0.68% - 1.61%	0.10%	0.22%	1.67%	complete

original: greedy problems test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256. main: greedy problems test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256. reward_only: greedy problems test 2048/2048, hard 1024/1024, probes 16/16; sampled complete problems 256/256.

Main minus control: 0.000 percentage points; paired 95% interval [-0.244, 0.244], on 2048 paired problems.

Comparison	Difference (pp)	Paired 95% interval (pp)	Problems
Main - reward-only (primary)	0.000	[-0.244, 0.244]	2048
main - original (secondary)	0.830	[0.438, 1.270]	2048
reward_only - original (secondary)	0.830	[0.391, 1.270]	2048

Comparisons with the original checkpoint are descriptive secondary estimates with unadjusted intervals.



Percentile bootstrap intervals can collapse at observed boundaries or when paired outcomes are identical. A zero-width interval does not prove zero population error or equivalence on unseen problems.

Probe performance, validity and resources

Condition	Training probe accuracy	Test validity	Hard validity	Cumulative train tokens	Stored eval tokens	Phase wall hours
original	0.00%	29.30%	10.64%	0	385121	0.000
main	0.00%	76.12%	2.05%	798675	217202	1.688
reward_only	0.00%	79.79%	3.52%	814802	229510	1.662

Within-problem expression diversity

Condition	Test greedy correct structures	Original-test sampled problems	Mean correct structures / original-test sampled problem
original	6	256	0.0156
main	23	256	0.0391
reward_only	23	256	0.0312

Greedy structure counts equal the number of successful test problems because only one answer is generated per problem. The sampled statistic counts distinct correct canonical ASTs within each problem before averaging, including zeros. It measures syntactic diversity, not algebraic or semantic equivalence.

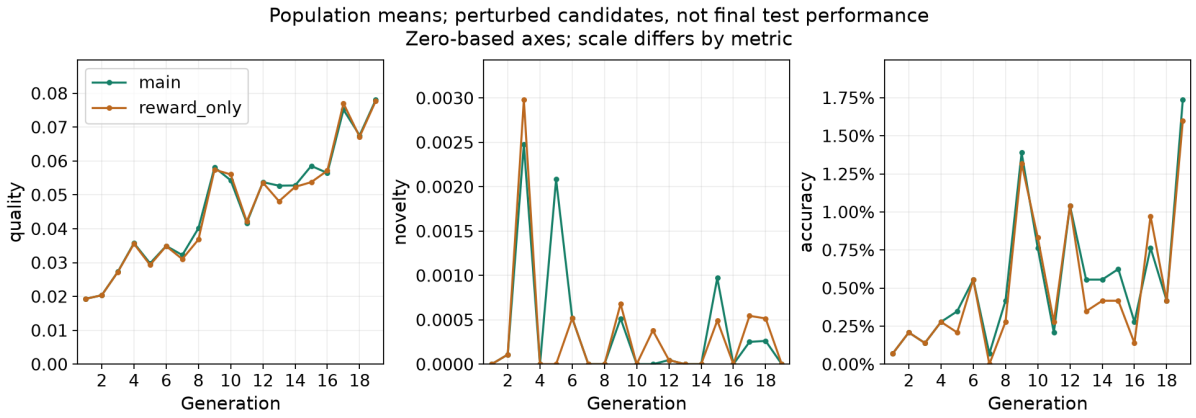
Train tokens count candidates in committed generations; evaluation tokens count persisted evaluation responses. Calibration, validation, interrupted batches and replay costs are excluded from these two table columns. Training hours are wall-clock phase duration and include checkpointing. Probes are training examples, not held-out data.

Did the novelty bonus affect optimization?

Descriptive diagnostic: compare the main arm's Z(Q) + 1.0 N with Z(Q) on its recorded populations. The control computes novelty but applies coefficient zero, so its novelty-induced weight changes are zero. This local comparison does not simulate the counterfactual training trajectory. main: positive novelty in 15 of 570 candidates; changed update weights in 9 of 19 generations (14 candidate weights). reward_only: positive novelty in 13 of 570 candidates; changed update weights in 0 of 19 generations (0 candidate weights).

Training dynamics and output inspection

Population metrics describe perturbed candidates, not the master checkpoint. Reward is the optimization signal; no cross-entropy loss is optimized in this pilot. The interactive result page shows the same first test problems across methods, including failures, without choosing examples by outcome.



Fixed example: original

test-00000 - legal, wrong target
Use the numbers [49, 11, 13] exactly once each to make 73. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: <answer>49 * (11 - 13)</answer>

Fixed example: main

test-00000 - legal, wrong target
Use the numbers [49, 11, 13] exactly once each to make 73. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: <answer> (13 - 49) * 11 </answer>

Fixed example: reward_only

test-00000 - invalid
Use the numbers [49, 11, 13] exactly once each to make 73. You may use +, -, *, / and parentheses. Do not use other numbers or operations. Return only <answer>expression</answer>.
Output: <answer> (49 - 11) * (11 + 13) </answer>

Implications and discussion

Both trained checkpoints score higher than the original model on this test set; the main-versus-control comparison determines whether novelty adds a benefit. Both also produce more legal expressions on the three/four-number test problems. Legality alone does not mean the target was reached. Both have lower expression validity on the five-number test problems than the original model, so the validity improvement does not transfer uniformly. This pilot is a controlled objective intervention under one initialization/seed schedule. It can identify a conditional difference or a feasibility failure, but cannot establish a general advantage of novelty training. Any improvement must be checked across additional seeds and separated from probe memorization. A rise in novelty alone, or a selected qualitative example, is insufficient evidence.

Limitations

Training-seed sensitivity is unmeasured. Embedding similarity is a proxy and may reward irrelevant wording; canonical expression trees are an independent syntactic diagnostic, not full algebraic equivalence. Synthetic arithmetic does not measure general language quality. Similar pretraining patterns cannot be excluded. The fixed-probe objective can overfit. Small time budgets, laptop thermal variability and batch-level floating-point behavior limit extrapolation. CPU tests, report generation, audits and backups ran alongside training; elapsed times are descriptive, not isolated systems benchmarks. Secondary endpoints are descriptive; no multiple-comparison significance claims are made. Generation-zero tensors and the complete historical embedding/archive trajectory were not retained. Historical initialization audits cannot replace these missing inputs for independent recomputation. The current verification checks raw grades, schedules, coefficients and final exports; it does not reconstruct every historical novelty distance.

Conclusion

The directional improvement hypothesis was not supported in this seed: main accuracy did not exceed reward-only ES. The paired interval includes zero, so this does not establish a performance difference. Multi-seed studies and archive ablations require a separately frozen amendment after this result is published.

Appendix: reproducibility and failures

Restore both pinned evidence releases and their separately published model exports using docs/continuation.md. The verify command recomputes evidence checks without inference; report-all regenerates both releases in a separate output tree. Both require explicit input paths. Existing training allowances have expired; these corrections execute no training or new model evaluation. Atomic checkpoints contain weights, all RNG streams, archive, cursor, config and deadline. Recorded technical failures: 1. Model-execution elapsed seconds: 20151.339. Model-execution timing is separate from later auditing and publication; the registration permits publication transfers to finish later. Exact exceptions and incomplete conditions are retained in the generated Markdown report and JSON artifacts.

Execution provenance: NVIDIA GeForce RTX 5070 Laptop GPU; Python 3.12.13. Initial execution revision: ddcd928c1392fa2eb5beaff5b7b6f192dfa04817. Training implementation revision: 9f8d30ef0103e3532ebefeafe7f55b8f8d163a80. Frozen generations per arm: 19; inference batch: 16. The timing rule uses the slowest calibration candidate, a complete update and checkpoint, and amortized validation cost. The budget reserves a fraction of each training phase and freezes the common count before training.

Embedding revision: 1110a243fdf4706b3f48f1d95db1a4f5529b4d41. ExTra upstream revision: ea53e68036356f8f276e261e886fbbc102633be3. Complete dependency versions, dataset hashes, absolute deadline, per-phase timings, token-work accounting, raw outputs and completion flags are machine-readable in the published experiment artifacts. Full-weight noise is regenerated by a hash of perturbation seed and tensor name using an explicit PyTorch generator. Bit-level replay is scoped to the recorded hardware and software environment.

The observed storage-failure recovery regenerated a batch of 16 sampled responses. The historical audit reports that decoded text, token IDs, identities and verifier fields matched the preserved failed-write snapshot; only elapsed batch time was excluded. The source hash and machine-readable check are published with the artifacts. This is evidence for this evaluation replay, not arbitrary training restarts or environments.

Empirical tables, figures, fixed-order examples and resource counts derive from the immutable raw evidence release linked below, through reporting.py. The dated correction report records the independent reanalysis and its limits.

References

1. ExTra
2. Evolution Strategies at Scale
3. Beyond the Best Guess
4. Novelty-seeking ES
5. EVOL-RL
6. DARLING
7. AC/DC
8. Evaluating Large Language Models Trained on Code
9. Pinned MiniLM input limit
10. Python expression syntax

Dated corrections and evidence limits

Immutable raw responses, configurations and logs

Research protocol and source

AI statement. Codex wrote this codebase and paper, and human cleanup and rewriting are in progress. A human provided the exact specification that was executed.